

# Area, Perimeter and Volume

GCSE Maths · Practice Questions

20 questions · 62 marks total

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## Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Work out the area of a rectangle measuring 7 cm by 5 cm.

.....

2 [2 marks]

Work out the area of a triangle with base 12 cm and perpendicular height 5 cm.

.....

3 [2 marks]

Work out the area of a trapezium with parallel sides 6 cm and 10 cm and perpendicular height 4 cm.

.....

4 [2 marks]

Work out the circumference of a circle of radius 7 cm, correct to 1 decimal place.

.....

5 [3 marks]

Work out the area of a circle of radius 6 cm, correct to 1 decimal place.

.....

6 [3 marks]

A cuboid measures 5 cm by 4 cm by 9 cm.

(a) Work out its volume.

(b) Work out its total surface area.

.....

7

[3 marks]

Work out the volume of a cylinder with radius 3 cm and height 8 cm, correct to 1 decimal place.

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8

[3 marks]

An L-shape is made by taking a rectangle 10 cm by 6 cm and cutting away a rectangle 4 cm by 3 cm from one corner.

Work out the area of the L-shape.

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9

[4 marks]

A semicircle has radius 5 cm.

(a) Work out its area, correct to 1 decimal place.

(b) Work out its perimeter, correct to 1 decimal place.

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10

[4 marks]

A prism has a trapezium cross-section with parallel sides 5 cm and 9 cm and perpendicular height 4 cm. The prism is 12 cm long.

(a) Work out the area of the cross-section.

(b) Work out the volume of the prism.

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## Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Work out the volume of a sphere of radius 3 cm. Give your answer in terms of  $\pi$ .

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2 [2 marks]

Work out the volume of a cone with radius 4 cm and height 9 cm. Give your answer in terms of  $\pi$ .

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3 [3 marks]

Work out the surface area of a sphere of radius 5 cm, correct to 3 significant figures.

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4 [3 marks]

A sector of a circle has radius 8 cm and angle  $45^\circ$ .

Work out the arc length, correct to 3 significant figures.

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5 [4 marks]

A sector of a circle has radius 10 cm and angle  $72^\circ$ .

(a) Work out its area, correct to 1 decimal place.

(b) Work out its perimeter, correct to 1 decimal place.

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6 [4 marks]

A cone has radius 5 cm and slant height 13 cm.

(a) Work out the curved surface area, correct to 3 significant figures.

(b) Work out the total surface area, correct to 3 significant figures.

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7 [4 marks]

A solid is made from a cylinder of radius 4 cm and height 10 cm, with a hemisphere of radius 4 cm on top.

Work out the total volume, correct to 3 significant figures.

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8

[4 marks]

A cylinder has volume  $500 \text{ cm}^3$  and height 10 cm.  
Work out its radius, correct to 2 decimal places.

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9

[4 marks]

A sphere of radius  $r$  fits exactly inside a cylinder of radius  $r$  and height  $2r$ .  
Show that the volume of the sphere is  $\frac{2}{3}$  of the volume of the cylinder.

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10

[5 marks]

A solid is made from a cone of radius 6 cm and slant height 10 cm standing on top of a cylinder of radius 6 cm and height 15 cm.

- (a) Work out the vertical height of the cone.
- (b) Work out the total volume of the solid, to the nearest  $\text{cm}^3$ .
- (c) Work out the total surface area of the solid, to 3 significant figures. (The solid has a flat circular base but no join surfaces.)

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